

MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

September 21, 2026

Outline

- 1 Real Analysis Lecture 8
 - Compactness

Outline

- 1 Real Analysis Lecture 8
 - Compactness

Warm-up Challenge

Problem

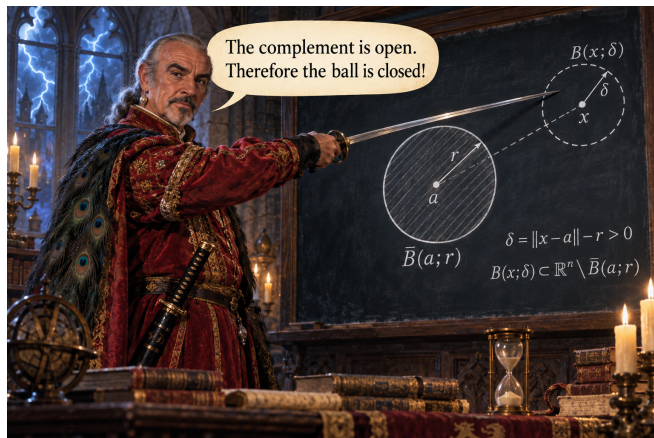
Let $\vec{a} \in \mathbb{R}^n$.

Prove that the closed ball

$$\overline{B}(\vec{a}; r) = \{\vec{x} \in \mathbb{R}^n : |\vec{x} - \vec{a}| \leq r\}$$

is closed.

Compact



Point set topology

We have been studying the beginnings of **point set topology**.

Point set topology

We have been studying the beginnings of **point set topology**. In topology, we start with a set X called the **space** and a collection of special subsets called **open sets**.

Point set topology

We have been studying the beginnings of **point set topology**. In topology, we start with a set X called the **space** and a collection of special subsets called **open sets**.

Fundamental questions:

Point set topology

We have been studying the beginnings of **point set topology**. In topology, we start with a set X called the **space** and a collection of special subsets called **open sets**.

Fundamental questions:

- kinds of points: interior points, boundary points, accumulation points, adherent points

Point set topology

We have been studying the beginnings of **point set topology**. In topology, we start with a set X called the **space** and a collection of special subsets called **open sets**.

Fundamental questions:

- kinds of points: interior points, boundary points, accumulation points, adherent points
- kinds of sets: open, closed, connected, component, compact

Point set topology

We have been studying the beginnings of **point set topology**. In topology, we start with a set X called the **space** and a collection of special subsets called **open sets**.

Fundamental questions:

- kinds of points: interior points, boundary points, accumulation points, adherent points
- kinds of sets: open, closed, connected, component, compact
- kinds of functions: continuous, homeomorphism

Point set topology

We have been studying the beginnings of **point set topology**. In topology, we start with a set X called the **space** and a collection of special subsets called **open sets**.

Fundamental questions:

- kinds of points: interior points, boundary points, accumulation points, adherent points
- kinds of sets: open, closed, connected, component, compact
- kinds of functions: continuous, homeomorphism

Answers must be in terms of open sets

Open covers

Definition

A **open cover** of a set $A \subseteq \mathbb{R}^n$ is a family $\{U_i : i \in I\}$ of open sets with $A \subseteq \bigcup_{i \in I} U_i$.

Open covers

Definition

A **open cover** of a set $A \subseteq \mathbb{R}^n$ is a family $\{U_i : i \in I\}$ of open sets with $A \subseteq \bigcup_{i \in I} U_i$.

A **subcover** is a family $\{U_j : j \in J\}$ with $J \subseteq I$ which still covers A , ie. $A \subseteq \bigcup_{j \in J} U_j$.

Open covers

Definition

A **open cover** of a set $A \subseteq \mathbb{R}^n$ is a family $\{U_i : i \in I\}$ of open sets with $A \subseteq \bigcup_{i \in I} U_i$.

A **subcover** is a family $\{U_j : j \in J\}$ with $J \subseteq I$ which still covers A , ie. $A \subseteq \bigcup_{j \in J} U_j$.

Examples:

Open covers

Definition

A **open cover** of a set $A \subseteq \mathbb{R}^n$ is a family $\{U_i : i \in I\}$ of open sets with $A \subseteq \bigcup_{i \in I} U_i$.

A **subcover** is a family $\{U_j : j \in J\}$ with $J \subseteq I$ which still covers A , ie. $A \subseteq \bigcup_{j \in J} U_j$.

Examples:

- $\{\mathbb{R}\}$ is an open cover of the interval $(0, 1]$

Open covers

Definition

A **open cover** of a set $A \subseteq \mathbb{R}^n$ is a family $\{U_i : i \in I\}$ of open sets with $A \subseteq \bigcup_{i \in I} U_i$.

A **subcover** is a family $\{U_j : j \in J\}$ with $J \subseteq I$ which still covers A , ie. $A \subseteq \bigcup_{j \in J} U_j$.

Examples:

- $\{\mathbb{R}\}$ is an open cover of the interval $(0, 1]$
- $\{(0, 1/2), (1/3, 2)\}$ is an open cover of the interval $(0, 1]$

Open covers

Definition

A **open cover** of a set $A \subseteq \mathbb{R}^n$ is a family $\{U_i : i \in I\}$ of open sets with $A \subseteq \bigcup_{i \in I} U_i$.

A **subcover** is a family $\{U_j : j \in J\}$ with $J \subseteq I$ which still covers A , ie. $A \subseteq \bigcup_{j \in J} U_j$.

Examples:

- $\{\mathbb{R}\}$ is an open cover of the interval $(0, 1]$
- $\{(0, 1/2), (1/3, 2)\}$ is an open cover of the interval $(0, 1]$
- $\{(\frac{1}{n}, \frac{n+1}{n}) : n \in \mathbb{Z}_+\}$ is an open cover of $(0, 1]$

Lindelöf Covering Theorem

Theorem (Lindelöf Covering Theorem)

Let $A \subseteq \mathbb{R}^n$ be a set and suppose $\{U_i : i \in I\}$ is an open covering of A . Then there exists a countable subcover $\{U_j : j \in J\}$.

Lindelöf Covering Theorem

Theorem (Lindelöf Covering Theorem)

Let $A \subseteq \mathbb{R}^n$ be a set and suppose $\{U_i : i \in I\}$ is an open covering of A . Then there exists a countable subcover $\{U_j : j \in J\}$.

Question

Can we do better than this?

Lindelöf Covering Theorem

Theorem (Lindelöf Covering Theorem)

Let $A \subseteq \mathbb{R}^n$ be a set and suppose $\{U_i : i \in I\}$ is an open covering of A . Then there exists a countable subcover $\{U_j : j \in J\}$.

Question

Can we do better than this?

Definition

A set $A \subseteq \mathbb{R}^n$ is called **compact** if every open cover of A has a *finite* subcover.

Compact



Challenge

Problem

Show that a singleton set $A = \{\vec{x}\}$ is compact.

Challenge

Problem

Show that a singleton set $A = \{\vec{x}\}$ is compact.

Solution

Challenge

Problem

Show that a singleton set $A = \{\vec{x}\}$ is compact.

Solution

Suppose that $\{U_i : i \in I\}$ is an open cover of X .

Challenge

Problem

Show that a singleton set $A = \{\vec{x}\}$ is compact.

Solution

Suppose that $\{U_i : i \in I\}$ is an open cover of X .
Then $A \subseteq \bigcup_{i \in I} U_i$.

Challenge

Problem

Show that a singleton set $A = \{\vec{x}\}$ is compact.

Solution

Suppose that $\{U_i : i \in I\}$ is an open cover of X .

Then $A \subseteq \bigcup_{i \in I} U_i$.

This means that $\vec{x} \in \bigcup_{i \in I} U_i$.

Challenge

Problem

Show that a singleton set $A = \{\vec{x}\}$ is compact.

Solution

Suppose that $\{U_i : i \in I\}$ is an open cover of X .

Then $A \subseteq \bigcup_{i \in I} U_i$.

This means that $\vec{x} \in \bigcup_{i \in I} U_i$.

Therefore $\vec{x} \in U_j$ for some $j \in I$.

Challenge

Problem

Show that a singleton set $A = \{\vec{x}\}$ is compact.

Solution

Suppose that $\{U_i : i \in I\}$ is an open cover of X .

Then $A \subseteq \bigcup_{i \in I} U_i$.

This means that $\vec{x} \in \bigcup_{i \in I} U_i$.

Therefore $\vec{x} \in U_j$ for some $j \in I$.

It follows that $\{U_j\}$ is a subcover of A consisting of a single set.

Challenge

Problem

Show that a singleton set $A = \{\vec{x}\}$ is compact.

Solution

Suppose that $\{U_i : i \in I\}$ is an open cover of X .

Then $A \subseteq \bigcup_{i \in I} U_i$.

This means that $\vec{x} \in \bigcup_{i \in I} U_i$.

Therefore $\vec{x} \in U_j$ for some $j \in I$.

It follows that $\{U_j\}$ is a subcover of A consisting of a single set.

Thus every open cover has a finite subcover, making A compact!

Bounded sets

Bounded sets

We will show that the compact subsets of $\mathbb{R}^n$ are exactly the closed sets that are bounded.

Bounded sets

We will show that the compact subsets of $\mathbb{R}^n$ are exactly the closed sets that are bounded.

Definition

A subset $A \subseteq \mathbb{R}^n$ is called **bounded** if there exists $\vec{x} \in \mathbb{R}^n$ and $r > 0$ such that $A \subseteq B(\vec{x}; r)$.

Bounded sets

We will show that the compact subsets of $\mathbb{R}^n$ are exactly the closed sets that are bounded.

Definition

A subset $A \subseteq \mathbb{R}^n$ is called **bounded** if there exists $\vec{x} \in \mathbb{R}^n$ and $r > 0$ such that $A \subseteq B(\vec{x}; r)$.

Theorem

If A is a compact set, then A must be bounded.

Bounded sets

We will show that the compact subsets of $\mathbb{R}^n$ are exactly the closed sets that are bounded.

Definition

A subset $A \subseteq \mathbb{R}^n$ is called **bounded** if there exists $\vec{x} \in \mathbb{R}^n$ and $r > 0$ such that $A \subseteq B(\vec{x}; r)$.

Theorem

If A is a compact set, then A must be bounded.

Proof.

Suppose A is a compact set.

Bounded sets

We will show that the compact subsets of $\mathbb{R}^n$ are exactly the closed sets that are bounded.

Definition

A subset $A \subseteq \mathbb{R}^n$ is called **bounded** if there exists $\vec{x} \in \mathbb{R}^n$ and $r > 0$ such that $A \subseteq B(\vec{x}; r)$.

Theorem

If A is a compact set, then A must be bounded.

Proof.

Suppose A is a compact set.

Consider the family of sets $\{U_i : i \in I\}$ where $I = \mathbb{Z}_+$ and $U_i = B(\vec{0}; i)$.



Challenge

Problem

Show that $\{U_i : i \in I\}$ is an open cover of A .

Challenge

Problem

Show that $\{U_i : i \in I\}$ is an open cover of A .

Solution

Note that U_i is an open subset for all i , so we need only show $A \subseteq \bigcup_{i \in I} U_i$.

Challenge

Problem

Show that $\{U_i : i \in I\}$ is an open cover of A .

Solution

Note that U_i is an open subset for all i , so we need only show $A \subseteq \bigcup_{i \in I} U_i$.

$$\bigcup_{i \in I} U_i = \bigcup_{i=1}^{\infty} B(\vec{0}; i) \subseteq \mathbb{R}^n.$$

Challenge

Problem

Show that $\{U_i : i \in I\}$ is an open cover of A .

Solution

Note that U_i is an open subset for all i , so we need only show $A \subseteq \bigcup_{i \in I} U_i$.

$$\bigcup_{i \in I} U_i = \bigcup_{i=1}^{\infty} B(\vec{0}; i) \subseteq \mathbb{R}^n.$$

Moreover, if $\vec{x} \in \mathbb{R}^n$, then we can choose $N \in \mathbb{Z}_+$ such that $N > |\vec{x}|$.

Challenge

Problem

Show that $\{U_i : i \in I\}$ is an open cover of A .

Solution

Note that U_i is an open subset for all i , so we need only show $A \subseteq \bigcup_{i \in I} U_i$.

$$\bigcup_{i \in I} U_i = \bigcup_{i=1}^{\infty} B(\vec{0}; i) \subseteq \mathbb{R}^n.$$

Moreover, if $\vec{x} \in \mathbb{R}^n$, then we can choose $N \in \mathbb{Z}_+$ such that $N > |\vec{x}|$.

Therefore $\vec{x} \in B(\vec{0}; N)$ and it follows $\vec{x} \in \bigcup_{i \in I} U_i$.

Challenge

Problem

Show that $\{U_i : i \in I\}$ is an open cover of A .

Solution

Note that U_i is an open subset for all i , so we need only show $A \subseteq \bigcup_{i \in I} U_i$.

$$\bigcup_{i \in I} U_i = \bigcup_{i=1}^{\infty} B(\vec{0}; i) \subseteq \mathbb{R}^n.$$

Moreover, if $\vec{x} \in \mathbb{R}^n$, then we can choose $N \in \mathbb{Z}_+$ such that $N > |\vec{x}|$.

Therefore $\vec{x} \in B(\vec{0}; N)$ and it follows $\vec{x} \in \bigcup_{i \in I} U_i$.

Thus $\mathbb{R}^n = \bigcup_{i \in I} U_i$, and since $A \subseteq \mathbb{R}^n$, we see $A \subseteq \bigcup_{i \in I} U_i$.

Challenge

Problem

Explain what it means that $\{U_i : i \in I\}$ has a finite subcover.

Challenge

Problem

Explain what it means that $\{U_i : i \in I\}$ has a finite subcover.

Solution

There is a finite subset $\{i_1, i_2, \dots, i_m\} \subseteq I$ with $A \subseteq \bigcup_{k=1}^m U_{i_k}$.

Challenge

Problem

Show that $\bigcup_{k=1}^m U_{i_k}$ is an open ball, and thus A is bounded.

Challenge

Problem

Show that $\bigcup_{k=1}^m U_{i_k}$ is an open ball, and thus A is bounded.

Solution

Without loss of generality, $i_1 < i_2 < \dots i_m$.

Challenge

Problem

Show that $\bigcup_{k=1}^m U_{i_k}$ is an open ball, and thus A is bounded.

Solution

Without loss of generality, $i_1 < i_2 < \dots i_m$. Then

$$B(\vec{0}; i_1) \subseteq B(\vec{0}; i_2) \subseteq \dots \subseteq B(\vec{0}; i_m).$$

Challenge

Problem

Show that $\bigcup_{k=1}^m U_{i_k}$ is an open ball, and thus A is bounded.

Solution

Without loss of generality, $i_1 < i_2 < \dots i_m$. Then

$$B(\vec{0}; i_1) \subseteq B(\vec{0}; i_2) \subseteq \dots \subseteq B(\vec{0}; i_m).$$

This means

$$A \subseteq \bigcup_{k=1}^m A_{i_k} = \bigcup_{k=1}^m B(\vec{0}; i_k) = B(\vec{0}, i_m).$$

Challenge

Problem

Show that $\bigcup_{k=1}^m U_{i_k}$ is an open ball, and thus A is bounded.

Solution

Without loss of generality, $i_1 < i_2 < \dots < i_m$. Then

$$B(\vec{0}; i_1) \subseteq B(\vec{0}; i_2) \subseteq \dots \subseteq B(\vec{0}; i_m).$$

This means

$$A \subseteq \bigcup_{k=1}^m A_{i_k} = \bigcup_{k=1}^m B(\vec{0}; i_k) = B(\vec{0}, i_m).$$

This shows A is bounded.

Compact implies closed

Theorem

Suppose that $A \subseteq \mathbb{R}^n$ is compact. Then A is closed.

Compact implies closed

Theorem

Suppose that $A \subseteq \mathbb{R}^n$ is compact. Then A is closed.

Proof.

Compact implies closed

Theorem

Suppose that $A \subseteq \mathbb{R}^n$ is compact. Then A is closed.

Proof.

Suppose that A is compact and let $\vec{x} \in \mathbb{R}^n \setminus A$.

Compact implies closed

Theorem

Suppose that $A \subseteq \mathbb{R}^n$ is compact. Then A is closed.

Proof.

Suppose that A is compact and let $\vec{x} \in \mathbb{R}^n \setminus A$.

We will show $\vec{x}$ cannot be an adherent point of A and therefore A contains all of its adherent points.

Compact implies closed

Theorem

Suppose that $A \subseteq \mathbb{R}^n$ is compact. Then A is closed.

Proof.

Suppose that A is compact and let $\vec{x} \in \mathbb{R}^n \setminus A$.

We will show $\vec{x}$ cannot be an adherent point of A and therefore A contains all of its adherent points.

Consider the family of sets $\{U_i : i \in I\}$ where $I = \mathbb{Z}_+$ and $U_i = \mathbb{R}^n \setminus \overline{B}(\vec{x}; 1/i)$. □

Challenge

Problem

Prove that $\bigcup_{i \in I} U_i = \mathbb{R}^n \setminus \{\vec{x}\}$.

Challenge

Problem

Explain why $\{U_i : i \in I\}$ is an open cover of A .

Challenge

Problem

Show that if $\{U_i : i \in I\}$ has a finite subcover, then $\vec{x}$ can't be an adherent point of $\vec{A}$.